



AEA South Pacific 2025

DZMx Plus with DZP_08 Controller

Flightcell



Flightcell SVS

Safety Voice Services for Air Traffic Communications



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Safety Voice Services for Air Traffic Communications

Trials will soon be underway in Northern FIR
Airservices Australia and Microflite
Standard Sat Calls in lieu of HF Radio
Commencing 7th October 2025

Airservices OneSky Communication upgrade starting 2028

DZMx Satellite only version and could have an upgrade path to SVS

ANP_00050 Cellular Antenna Replacing ANP_00033 Cellular Antenna

Flightcell

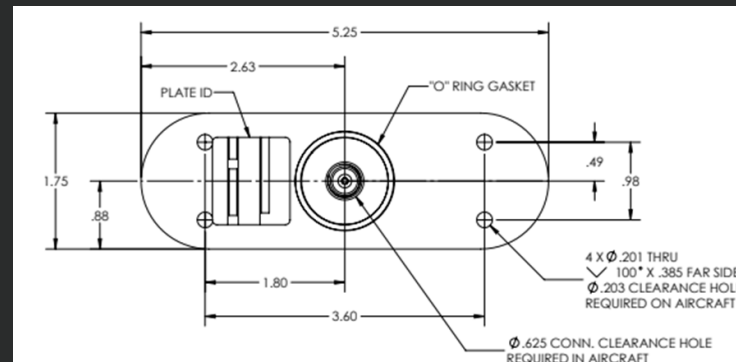


ANP_00050

ANP_00033

Identical Footprint

Mounting Area, Clearances, RF Connector and Mounting Holes



Why Change Then?

Lower Profile

Lighter

Greater Mechanical Integrity

Better Performance

MIMO - Multiple Input Multiple Output

4G 2x2 (2 inputs and 2 outputs)

5G minimum 4x4

3G Sunset Cell performance has changed

Diversity modem Multiple IN & 1 out

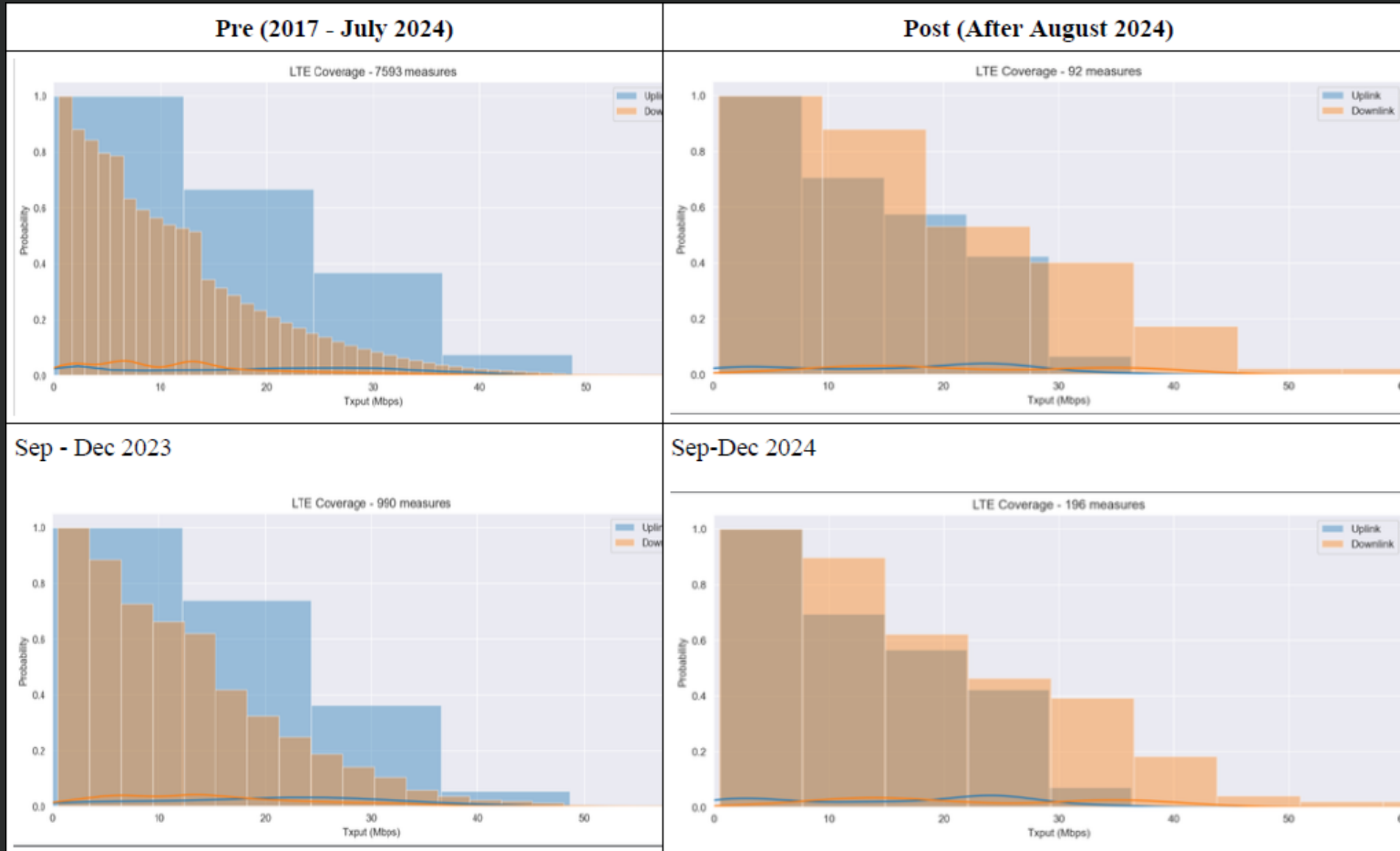
DZP_04-xxx-xxx Rev 8

DZP_07-xxx-xxx Rev 5

Due Release Q4 2025

Have a choice to use 1 or 2 antennas

DZMx and DZMx Plus Cell Diversity



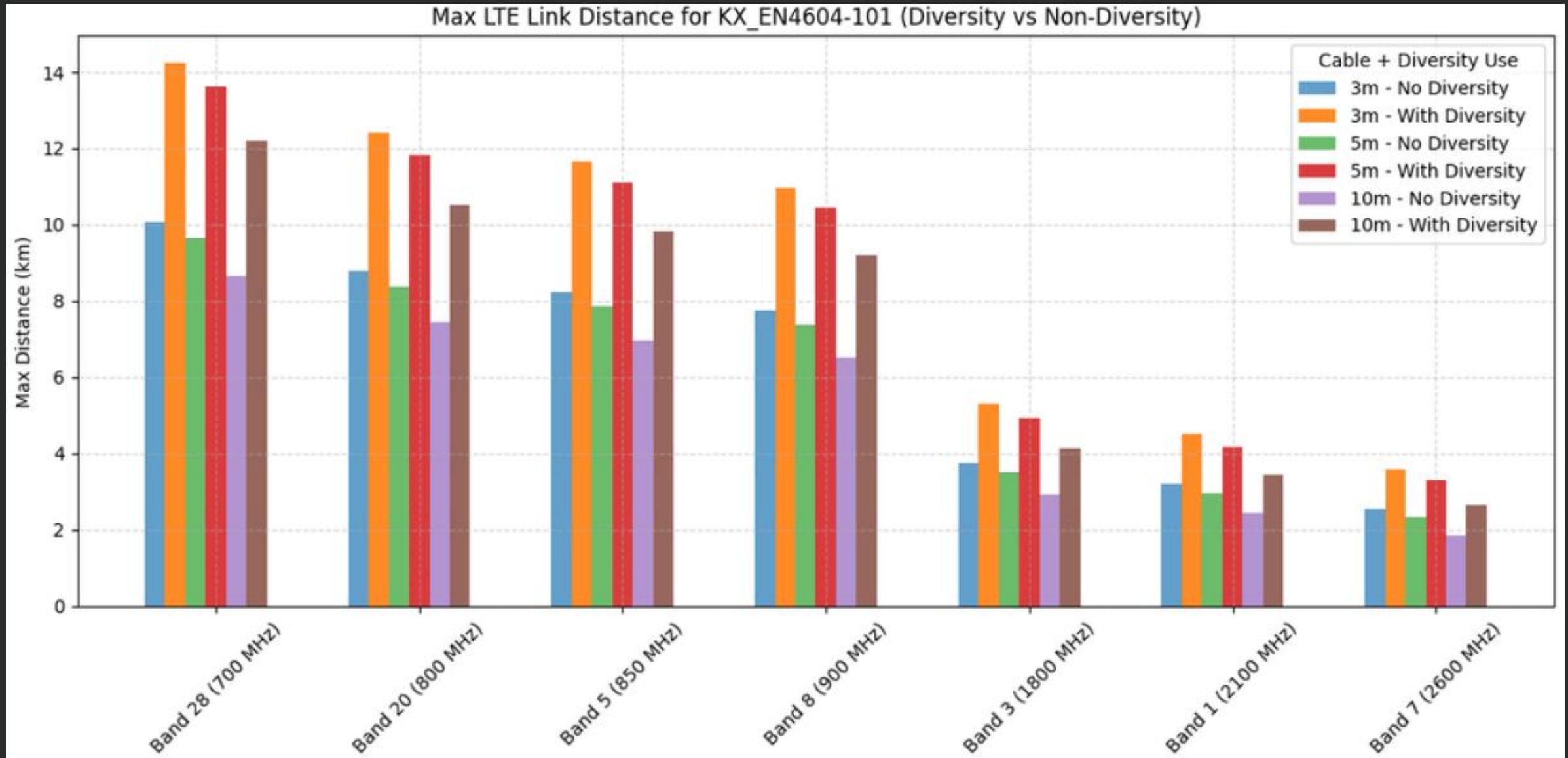
Coax for Cell Antenna Installation



Coax Cable Attenuation Comparison Table

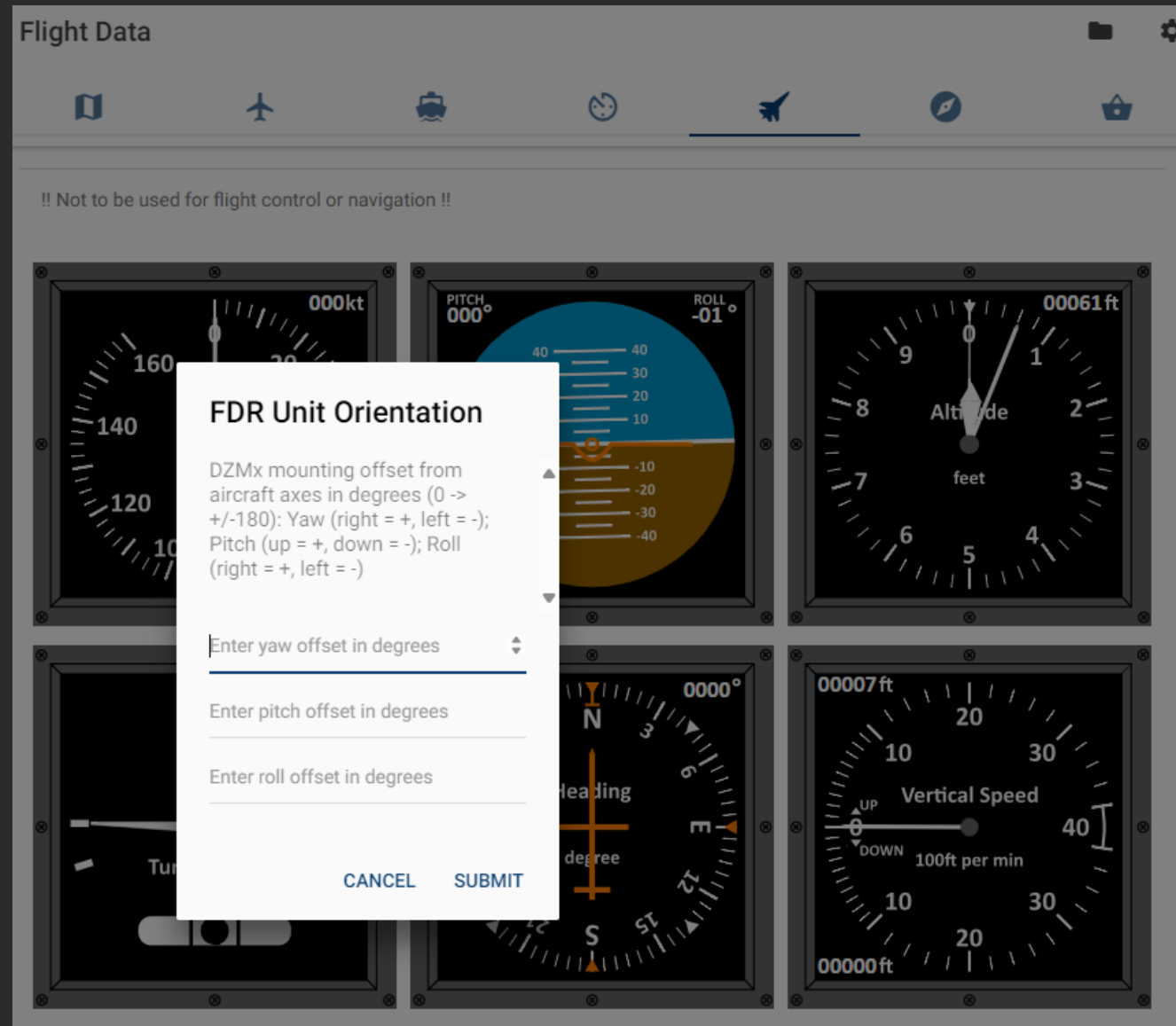
	Frequency	Attenuation dB/m	Antenna coax <u>DZMx</u> (5m)	Antenna coax <u>DZMx</u> Plus (7m)
S33141	0.4 GHz	0.154	0.77	1.078
	1.0 GHz	0.243	1.215	1.701
	1.6 GHz	0.312	1.56	2.184
	5.0 GHz	0.561	2.805	3.927
RG393	1.0 GHz	0.2461	1.2305	1.7227
	5.0 GHz	0.689	3.445	4.823
RG400	0.4 GHz	0.2953	1.4765	2.0671
	1.0 GHz	0.4823	2.4115	3.3761
	3.0 GHz	0.8825	4.4125	6.1775
	5.0 GHz	1.1844	5.922	8.2908
KX_EN4604-010 (<u>Huber+Suhner</u>)	0.6 GHz	0.177	0.885	1.239
	0.8 GHz	0.206	1.03	1.442
	1.0 GHz	0.231	1.155	1.617
	1.20 GHz	0.253	1.265	1.771
	1.40 GHz	0.274	1.37	1.918
	1.60 GHz	0.294	1.47	2.058
	1.80 GHz	0.313	1.565	2.191
	2.00 GHz	0.330	1.65	2.31
	4.00 GHz	0.475	2.375	3.325
	6.00 GHz	0.590	2.95	4.13

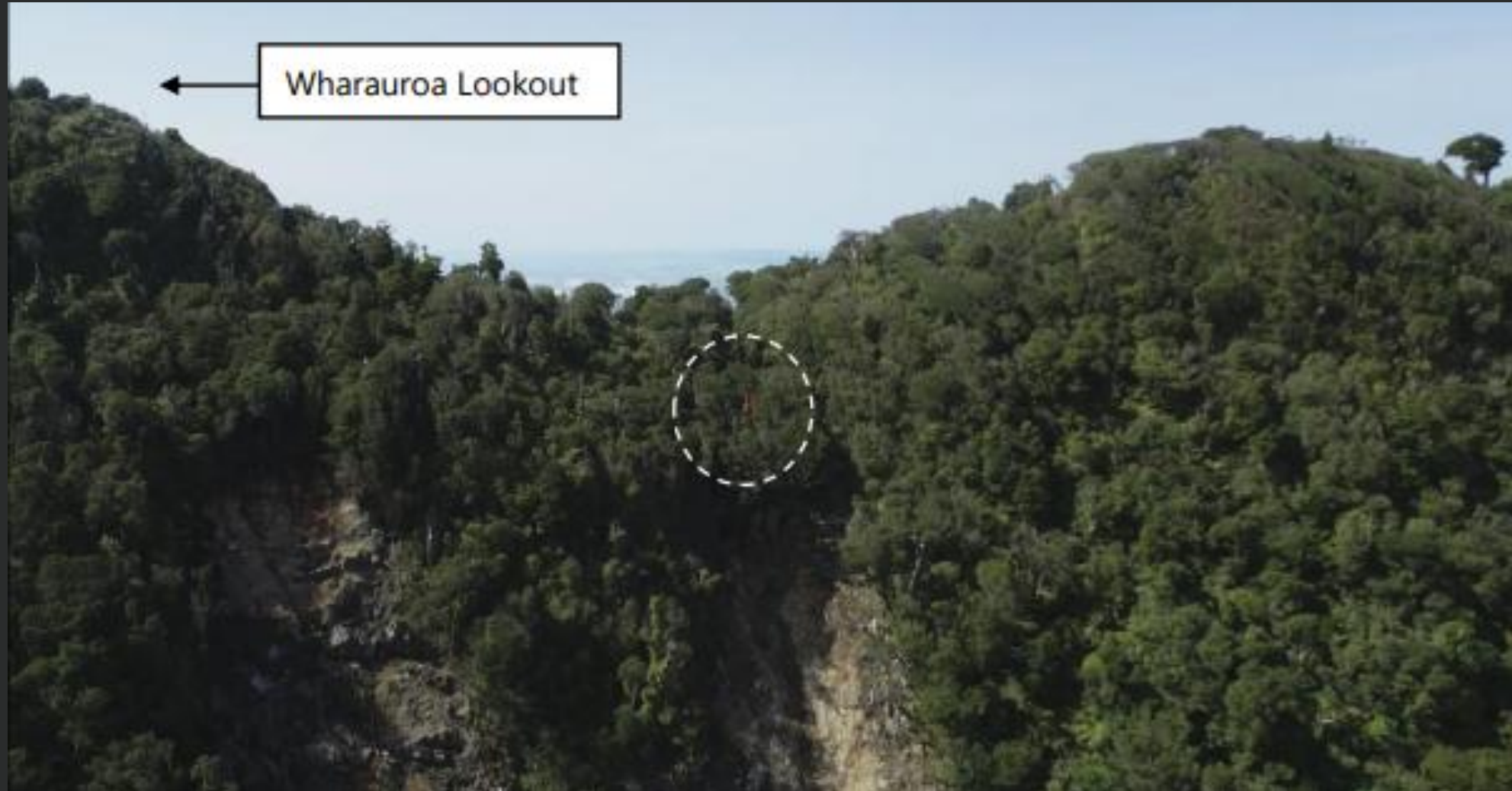
DZMx and DZMx Plus Cell Antenna Installation



Calibrate DZMx MPU

Flightcell



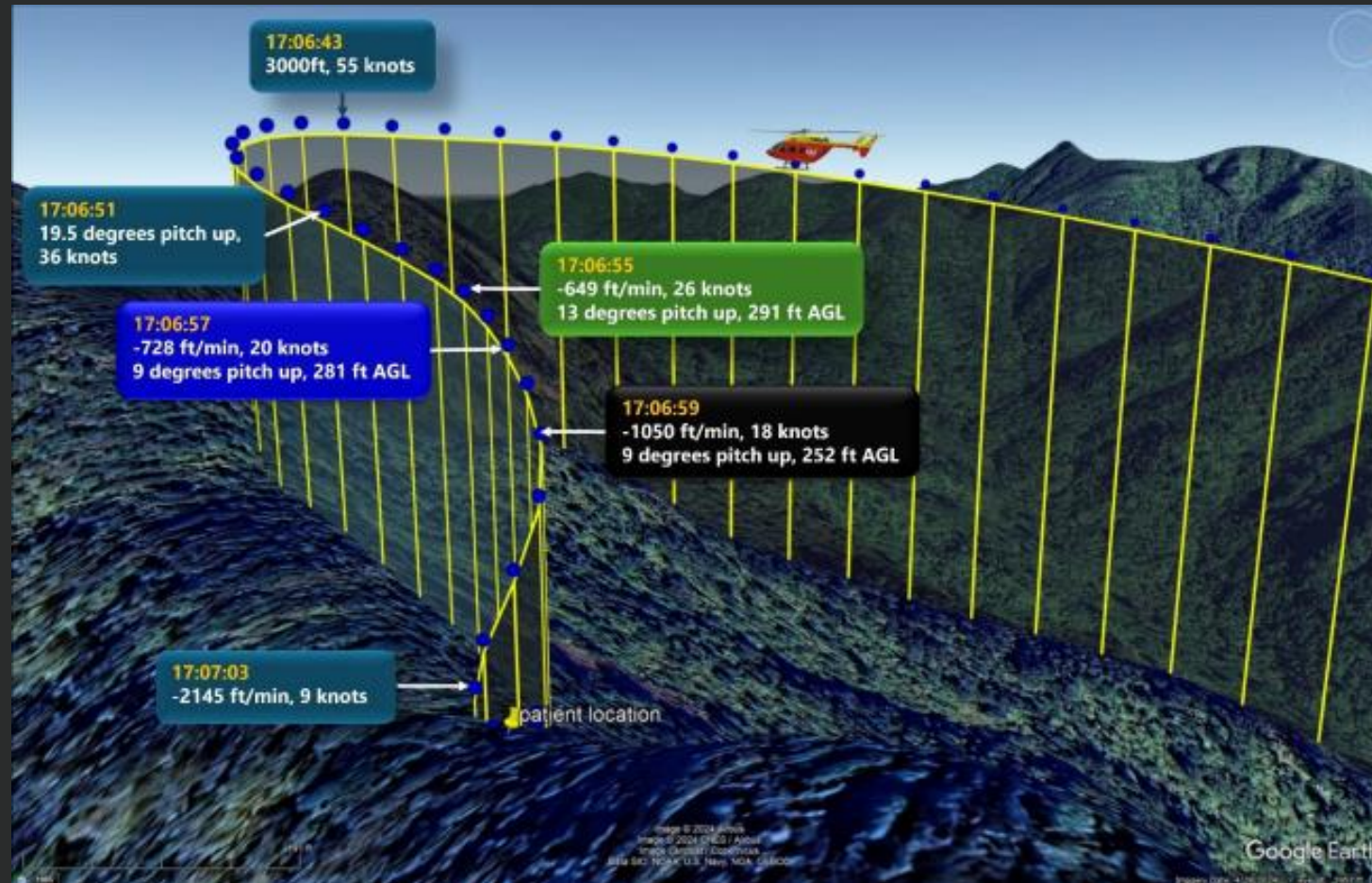


Accident Scene

Flightcell



Helicopter had entered Vortex Ring State - VRS



The helicopter's flight path was recreated using data from the Flightcell DZMx. The green data box marked 17:06:55 indicates where the helicopter entered VRS due to the combined effects of descent rate, wind, and pitch attitude – Image supplied by TAIC



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